

Design of Imbalanced Mach–Zehnder Interferometers for Group-Index Extraction

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Abstract—This proposal presents five imbalanced Mach–Zehnder interferometers (MZIs) designed to extract the group index of a 500×220 nm silicon-on-insulator strip waveguide for the quasi-TE mode. The path-length difference ΔL is varied from 86.4 to 786.8 μm , producing simulated free spectral ranges (FSRs) from approximately 6.88 to 0.75 nm near 1550 nm. Lumerical MODE simulations give an effective index $n_{\text{eff}} = 2.446$ and a group index $n_g = 4.054$ at 1550 nm. A quadratic compact model is fitted to the simulated effective index and used to calculate the MZI transmission spectra. After fabrication, the group index will be extracted from the measured fringe spacing and compared with the simulated value.

I. INTRODUCTION

THE group index n_g of a waveguide determines the free spectral range of resonators and interferometers and the delay of optical signals. An imbalanced MZI provides a direct way to measure it from the spacing between adjacent transmission fringes.

The objective of this design is to extract the group index of the standard strip waveguide from measured MZI spectra and compare it with the value obtained from waveguide simulation. Five MZIs with different path-length differences ΔL are included. For an ideal imbalanced MZI, increasing ΔL decreases the FSR, providing several devices with different fringe spacings for the same waveguide geometry.

II. THEORY

A. Waveguide compact model

The wavelength dependence of the effective index is represented by a second-order expansion around $\lambda_0 = 1.55 \mu\text{m}$ [1]:

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2. \quad (1)$$

The group index is defined as

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (2)$$

For the polynomial in (1),

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda [n_2 + 2n_3(\lambda - \lambda_0)]. \quad (3)$$

B. MZI transfer function

In each MZI, a Y-branch divides the input field between two waveguide arms [2]. The arm lengths are L_1 and

$$L_2 = L_1 + \Delta L, \quad (4)$$

where ΔL is the path-length difference.

The propagation constant is

$$\beta(\lambda) = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}. \quad (5)$$

The phase difference accumulated between the two arms is therefore

$$\phi(\lambda) = \beta(\lambda)\Delta L = \frac{2\pi n_{\text{eff}}(\lambda)\Delta L}{\lambda}. \quad (6)$$

Neglecting propagation loss, the normalized transmission for an MZI recombined by a second Y-branch, as used for MZI4 and MZI5, is

$$T_Y = \frac{1}{2} [1 + \cos(\beta\Delta L)]. \quad (7)$$

MZI1–MZI3 use a 2×2 3-dB coupler at the output. The two output ports have complementary transmission:

$$T_{1,2} = \frac{1}{2} [1 \pm \sin(\beta\Delta L)]. \quad (8)$$

C. Free spectral range and group-index extraction

Adjacent transmission minima or maxima occur when the interferometric phase changes by 2π . For a small wavelength interval corresponding to one FSR,

$$\left| \frac{d\phi}{d\lambda} \right| \text{FSR} \approx 2\pi. \quad (9)$$

Differentiating (6) gives

$$\frac{d\phi}{d\lambda} = 2\pi\Delta L \frac{d}{d\lambda} \left(\frac{n_{\text{eff}}}{\lambda} \right) \quad (10)$$

$$= \frac{2\pi\Delta L}{\lambda^2} \left(\lambda \frac{dn_{\text{eff}}}{d\lambda} - n_{\text{eff}} \right). \quad (11)$$

Using the definition of group index in (2),

$$\frac{d\phi}{d\lambda} = -\frac{2\pi n_g \Delta L}{\lambda^2}. \quad (12)$$

Substitution into (9) gives

$$\text{FSR} \approx \frac{\lambda^2}{n_g \Delta L}. \quad (13)$$

The group index can therefore be obtained from the measured fringe spacing using

$$n_g \approx \frac{\lambda^2}{\Delta L \cdot \text{FSR}}, \quad (14)$$

where λ is taken near the midpoint of the two adjacent fringes used to determine the FSR. Equation (14) assumes that the group index changes only slightly over one FSR.

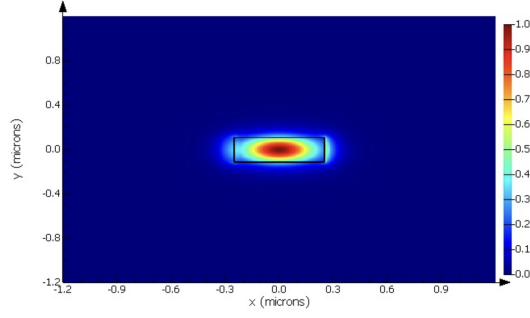


Fig. 1. Fundamental quasi-TE mode of the 500×220 nm strip waveguide at 1550 nm. The MODE simulation gives $n_{\text{eff}} = 2.4463$, an effective area of $0.190 \mu\text{m}^2$, and a TE polarization fraction of 98%.

III. MODELLING AND SIMULATION

A. Waveguide

The waveguide is a silicon strip with width $W = 500$ nm and height $H = 220$ nm on buried oxide with oxide cladding. The waveguide height is fixed by the fabrication process. Only the fundamental quasi-TE mode is considered, consistent with the TE grating couplers used in the layout.

The waveguide modes were calculated using the finite-difference eigenmode (FDE) solver in Lumerical MODE. The simulations used the Palik material models for Si and SiO_2 and a 2.5-nm mesh in both transverse directions. Figure 1 shows the fundamental quasi-TE mode at 1550 nm.

A wavelength sweep was performed at 10 points from 1550 to 1600 nm. Equation (1) was fitted to the simulated effective-index values. Figure 2 shows the MODE results and the fitted compact model.

With λ expressed in μm , the fitted model is

$$n_{\text{eff}}(\lambda) = 2.4463 - 1.0374(\lambda - 1.55) - 0.1329(\lambda - 1.55)^2. \quad (15)$$

The maximum residual of the fit over the simulated wavelength range is 9×10^{-6} . Using (3), the compact model gives $n_g = 4.054$ at 1550 nm and $n_g = 4.075$ at 1600 nm. These values agree with the group-index values obtained directly from the MODE sweep. Values below 1550 nm in the circuit calculation use an extrapolation of the fitted polynomial.

B. Design variations

The arm lengths of the five MZIs were measured from the layout in KLayout using SiEPIC-Tools [3]. The resulting path-length differences are listed in Table I.

The five values of ΔL produce a range of measurable fringe spacings. The largest simulated FSR is approximately 6.88 nm and the smallest simulated FSR is approximately 0.75 nm.

C. Circuit simulation

The transmission spectra were calculated using the wavelength-dependent effective index from (15) and the MZI transfer functions in (7) and (8).

Figure 3 shows the resulting transmission spectra. MZI1–MZI3 have two complementary output ports, while MZI4 and

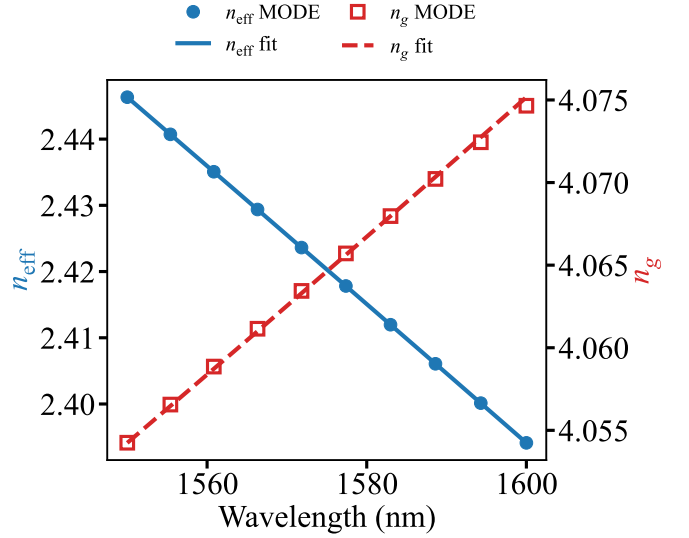


Fig. 2. Effective index and group index obtained from Lumerical MODE. The lines show the fitted effective-index model and the corresponding group index calculated from (3).

TABLE I
MZI DESIGN VARIATIONS AND SIMULATED FSR

Device	Combiner	ΔL (μm)	FSR (nm)
MZI4	Y-branch	86.417	6.88
MZI1	2×2	109.237	5.43
MZI2	2×2	148.957	3.98
MZI5	Y-branch	224.786	2.63
MZI3	2×2	786.767	0.75

MZI5 have a single monitored output. The FSR for each design was determined from the spacing between adjacent minima near 1550 nm.

The grating-coupler wavelength response is not included in these spectra. The purpose of this simulation is to compare the interferometric fringe spacing produced by the different values of ΔL . The final panel of Fig. 3 shows the expected inverse dependence of FSR on path-length difference.

D. Group-index extraction from simulated spectra

Equation (14) was checked using the simulated FSR. For MZI1, $\Delta L = 109.237 \mu\text{m}$ and the simulated FSR is 5.43 nm near 1550 nm. Substituting these values into (14) gives $n_g \approx 4.05$, close to the MODE value of 4.054.

E. Extraction from experimental data

After fabrication, the FSR will be measured from the spacing between adjacent minima in each MZI transmission spectrum. The group index will then be calculated using (14) and the corresponding path-length difference ΔL . The extracted group index will be compared with the MODE simulation and across the five MZI designs.

IV. FABRICATION

To be completed later.

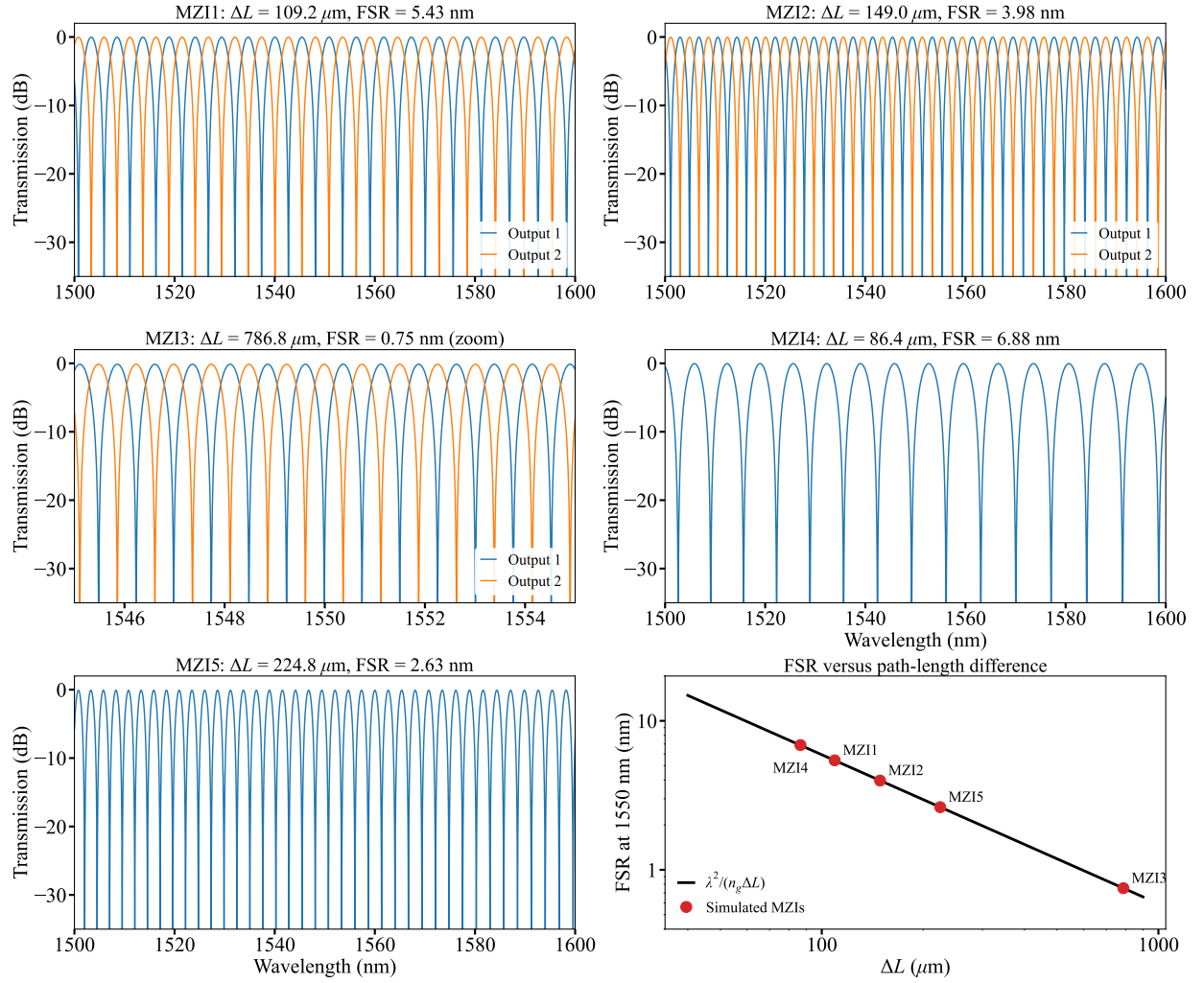


Fig. 3. Calculated transmission spectra of the five MZIs with the grating-coupler response excluded. MZI1–MZI3 have two complementary outputs. MZI3 is shown over 1545–1555 nm to make its smaller fringe spacing visible. The bottom-right panel compares the simulated FSR values with the dependence predicted by (13).

The submitted layout is shown in Fig. 4. Each MZI uses TE grating couplers placed on the standard 127- μm pitch for automated optical testing. MZI1–MZI3 use one input and two output grating couplers, while MZI4 and MZI5 use one input and one output grating coupler.

V. EXPERIMENTAL DATA

To be completed later.

VI. ANALYSIS

To be completed later.

VII. CONCLUSION

To be completed later.

REFERENCES

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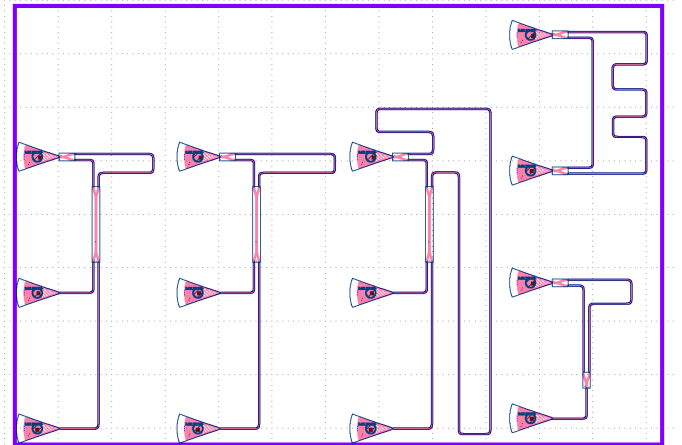


Fig. 4. Layout of the five MZIs in EBeam_vaddiyaswant.gds. Left to right: MZI1, MZI2, and MZI3; top right: MZI4; bottom right: MZI5.